

Task 1: Find the IP Address of the Device

Command used

- ipconfig

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix  . :  
IPv6 Address. . . . . : 2409:40c1:4027:22d4:319a:c6b:29b1:a4f6  
Temporary IPv6 Address. . . . . : 2409:40c1:4027:22d4:5815:b72c:db85:557a  
Link-local IPv6 Address . . . . . : fe80::da02:197f:18c5:84f9%14  
IPv4 Address. . . . . : 10.41.112.162  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::64e0:fdff:fe30:47e1%14  
                             10.41.112.126
```

Detail

Information	Answer
IP Address	10.41.112.162
Version	IPv4
Type	Private IP address

Task 2: Identify Private and Public IP Addresses

The standard private IPv4 ranges are:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

IP Address	Type	Reason
192.168.1.10	Private	Falls within 192.168.0.0/16.
10.0.0.5	Private	Falls within 10.0.0.0/8.

172.168.5.20	Private	Falls within 172.16.0.0/12.
8.8.8.8	Public	Does not belong to any RFC 1918 private range.
123.45.67.89	Public	Does not belong to any RFC 1918 private range.

Task 3: Python Script to Validate an IPv4 Address

```

ip = input("Enter an IPv4 address: ")
parts = ip.split(".")
valid = True
if len(parts) != 4:
    valid = False
else:
    for part in parts:
        if not part.isdigit():
            valid = False
            break
        number = int(part)
        if number < 0 or number > 255:
            valid = False
            break
if valid:
    print("Valid IPv4 address")
else:
    print("Invalid IPv4 address")

```

Task 4: Using IP Addresses to Detect Suspicious Login Activity

Example

Suppose a user normally logs into Instagram from:

Ahmedabad, India

with an IP address associated with that region.

Later, the account suddenly receives a login from an IP address associated with:

London, United Kingdom

within a very short period.

The service could consider this unusual because the two locations may be difficult to reach within that time.

Simplified Process

Normal Login



IP: 49.x.x.x



Location: India



Account activity looks normal



New Login



IP: 81.x.x.x



Location: United Kingdom



Unusual location detected



Additional security check

The service might respond by:

- Sending a security alert.
- Asking the user to verify the login.
- Requesting additional authentication.
- Temporarily restricting suspicious activity.